

FPORTAFOLIO

```
library(quantmod)
```

Cargando paquete requerido: xts

Cargando paquete requerido: zoo

Adjuntando el paquete: 'zoo'

The following objects are masked from 'package:base':

as.Date, as.Date.numeric

Cargando paquete requerido: TTR

Registered S3 method overwritten by 'quantmod':

method from
as.zoo.data.frame zoo

```
library(fPortfolio)
```

Cargando paquete requerido: timeDate

Cargando paquete requerido: timeSeries

Adjuntando el paquete: 'timeSeries'

The following object is masked from 'package:zoo':

```
time<-
```

The following objects are masked from 'package:graphics':

```
lines, points
```

Cargando paquete requerido: fBasics

Adjuntando el paquete: 'fBasics'

The following object is masked from 'package:TTR':

```
volatility
```

Cargando paquete requerido: fAssets

```
library(tidyverse)
```

```
-- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
v dplyr      1.1.4      v readr      2.1.5
v forcats    1.0.0      v stringr    1.5.1
v ggplot2    3.5.1      v tibble     3.2.1
v lubridate  1.9.4      v tidyr      1.3.1
v purrr      1.0.4
```

```
-- Conflicts ----- tidyverse_conflicts() --
x dplyr::filter() masks timeSeries::filter(), stats::filter()
x dplyr::first()  masks xts::first()
x dplyr::lag()    masks timeSeries::lag(), stats::lag()
x dplyr::last()   masks xts::last()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become
```

```
library(timeSeries)
```

```
precios <- c("COST", "MA", "PG", "DIS", "SPOT")
getSymbols(precios, from = "2019-01-02", periodicity = "daily")
```

```
[1] "COST" "MA" "PG" "DIS" "SPOT"
```

```
# Extraer y limpiar precios (sin NA ni ceros)
precios_list <- list(
  COST = na.omit(COST$COST.Close),
  MA    = na.omit(MA$MA.Close),
  PG    = na.omit(PG$PG.Close),
  DIS   = na.omit(DIS$DIS.Close),
  SPOT  = na.omit(SPOT$SPOT.Close)
)

# Calcular retornos logarítmicos
R.C <- diff(log(precios_list$COST)) * 100
R.M <- diff(log(precios_list$MA)) * 100
R.P <- diff(log(precios_list$PG)) * 100
R.D <- diff(log(precios_list$DIS)) * 100
R.S <- diff(log(precios_list$SPOT)) * 100

# Alinear longitudes (por si difieren)
retornos <- na.omit(merge(R.C, R.M, R.P, R.D, R.S))
colnames(retornos) <- c("R.C", "R.M", "R.P", "R.D", "R.S")

mu <- colMeans(retornos); mu
```

	R.C	R.M	R.P	R.D	R.S
	0.099810534	0.068728418	0.034220986	-0.001765793	0.108976913

```
Datos <- as.data.frame(retornos)
S <- cov(Datos); S
```

	R.C	R.M	R.P	R.D	R.S
R.C	2.1486098	1.200212	0.9140233	0.9237187	1.3515339
R.M	1.2002122	3.435939	1.0450716	2.2879038	1.9594761
R.P	0.9140233	1.045072	1.6699095	0.7710093	0.3741879
R.D	0.9237187	2.287904	0.7710093	4.2788822	2.2418133
R.S	1.3515339	1.959476	0.3741879	2.2418133	9.2172984

```
i <- matrix(1, 5, 1)
Si <- solve(S) %*% i
Su <- solve(S) %*% mu

A <- as.numeric(t(i) %*% solve(S) %*% i); A
```

```
[1] 0.7759972
```

```
B <- as.numeric(t(i) %*% solve(S) %*% mu);B
```

```
[1] 0.04079784
```

```
C <- as.numeric(t(mu) %*% solve(S) %*% mu);C
```

```
[1] 0.006198903
```

```
D <- A * C - B^2
```

```
mu.bar <- 0.25
```

```
W <- ((C - mu.bar * B) / D) * Si + ((mu.bar * A - B) / D) * Su; W
```

```
      [,1]  
R.C 1.8821501  
R.M 0.7946234  
R.P -0.8237985  
R.D -1.1619200  
R.S 0.3089450
```

```
# Riesgo del portafolio
```

```
s.W <- sqrt(t(W) %*% S %*% W); s.W
```

```
      [,1]  
[1,] 3.301989
```

```
# Portafolio global mínimo
```

```
w.g <- Si / A; w.g
```

```
      [,1]  
R.C 0.28429787  
R.M 0.00700651  
R.P 0.54924898  
R.D 0.11246421  
R.S 0.04698244
```

```
s.g <- 1 / sqrt(A); s.g
```

```
[1] 1.135194
```

```
mu.g <- t(w.g) %*% mu; mu.g
```

```
      [,1]  
[1,] 0.05257472
```

```
# Portafolio tangente
```

```
Rf <- 10 / 365
```

```
E <- as.numeric(t(mu - Rf * i) %*% solve(S) %*% (mu - Rf * i));E
```

```
[1] 0.004545876
```

```
sharpe <- sqrt(E); sharpe
```

```
[1] 0.06742311
```

```
w.tg <- t(solve(S) %*% (mu - Rf * i)) / as.numeric(t(mu - Rf * i) %*% solve(S) %*% i); w.tg
```

```
      R.C      R.M      R.P      R.D      R.S  
[1,] 1.963651 0.834797 -0.8938329 -1.226922 0.3223068
```

```
mu.tg <- w.tg %*% mu; mu.tg
```

```
      [,1]  
[1,] 0.26007
```

```
s.tg <- abs(mu.tg - Rf) / sharpe; s.tg
```

```
      [,1]  
[1,] 3.450934
```

```

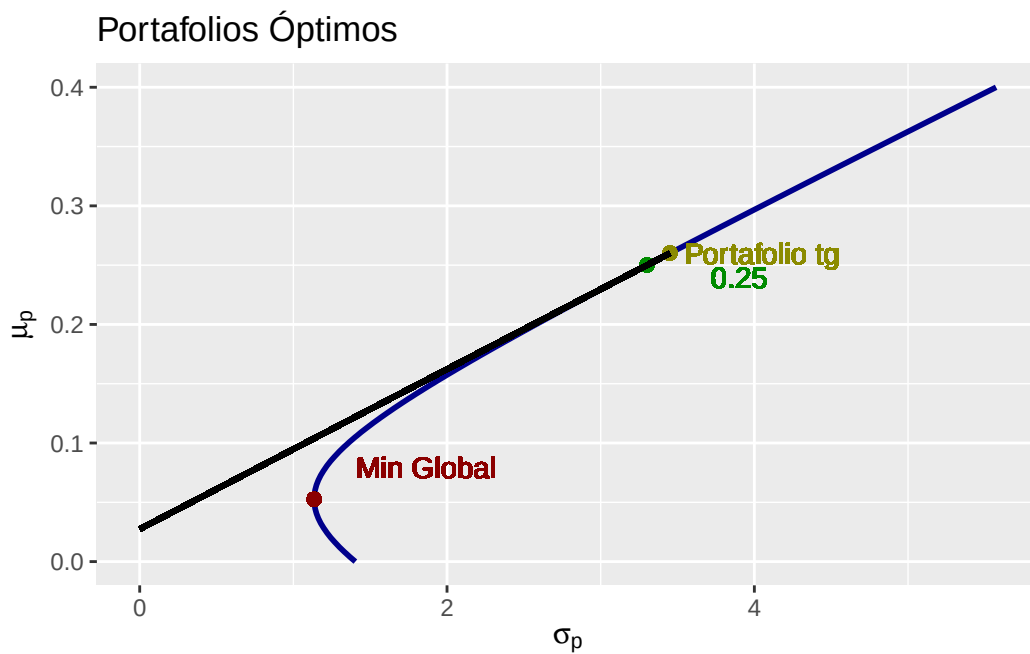
x<-seq(0,7, length=100)
y<-seq(0, 0.4, length=100)

df <- expand.grid(x=x, y=y) %>% mutate(z = D * x^2 - A * y^2 + 2 * B * y - C)

p<-ggplot(df, aes(x=x,y=y,z=z))+
  geom_contour(bins=0, lwd=1, colour="blue4")+
  labs(x=expression(sigma[p]),y=expression(mu[p]))+
  ggtitle("Portafolios Óptimos")

p+geom_point(aes(x=s.W, y=mu.bar), colour="green4", size = 2)+
  geom_point(aes(x=s.g, y=mu.g), colour="red4", size = 2)+
  geom_point(aes(x=s.tg, y=mu.tg), colour="yellow4", size = 2)+
  geom_text(aes(s.W + 0.6, y = mu.bar - 0.01, label="0.25"), color="green4")+
  geom_text(aes(x=s.g, y=mu.g, label="Min Global"), vjust=-1, hjust=-0.3, color="red4")+
  geom_text(aes(x=s.tg+0.6, y=mu.tg, label="Portafolio tg"), color="yellow4")+
  geom_segment(aes(x=0, xend=s.tg, y=Rf, yend=mu.tg), color="black", linewidth=1)

```



```

library(ggplot2)
i<-matrix(c(1,1,1,1,1),5)
mu <- rbind(mean(R.C[-1]), mean(R.M[-1]), mean(R.P[-1]), mean(R.D[-1]), mean(R.S[-1]));mu

```

```

      [,1]
[1,] 0.099810534
[2,] 0.068728418
[3,] 0.034220986
[4,] -0.001765793
[5,] 0.108976913

```

```
S <- cov(Datos); S
```

```

      R.C      R.M      R.P      R.D      R.S
R.C 2.1486098 1.200212 0.9140233 0.9237187 1.3515339
R.M 1.2002122 3.435939 1.0450716 2.2879038 1.9594761
R.P 0.9140233 1.045072 1.6699095 0.7710093 0.3741879
R.D 0.9237187 2.287904 0.7710093 4.2788822 2.2418133
R.S 1.3515339 1.959476 0.3741879 2.2418133 9.2172984

```

```

P<- matrix(runif(9000, min=0, max=100), ncol=5)
W=P/rowSums(P)

```

```

risk <- numeric(1800)
mu.p <- numeric(1800)

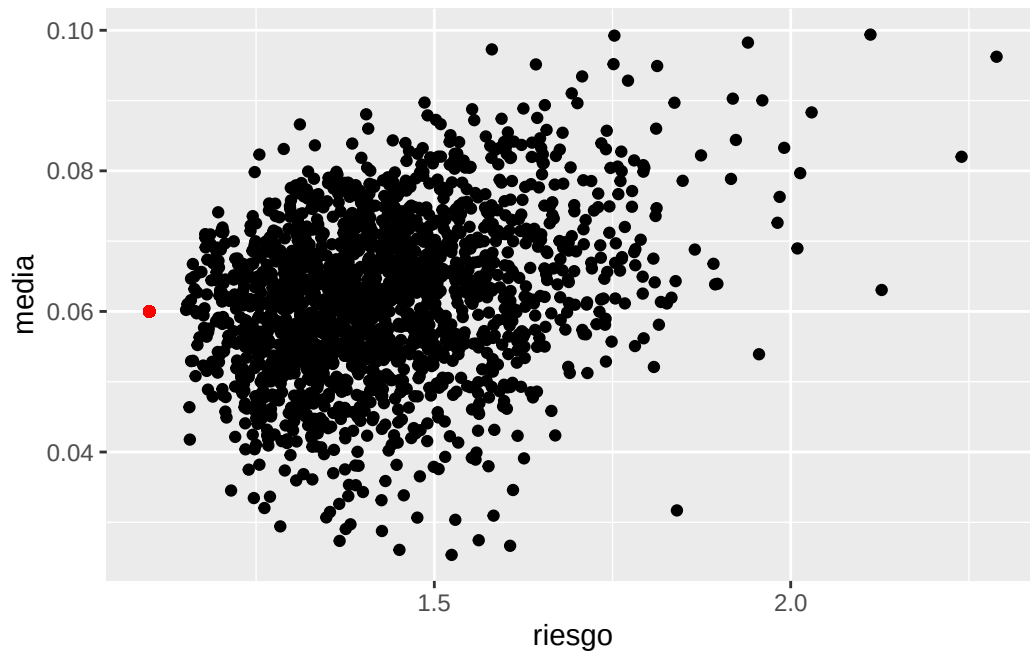
for(j in 1:1800){
  risk[j] <- sqrt(t(W[j,]) %*% S %*% W[j,])
  mu.p[j] <- sum(W[j,] * mu[,1])
}

```

```
datos <- data.frame(riesgo=risk, media=mu.p)
```

```
ggplot(datos, aes(x=riesgo, y=media))+geom_point()+geom_point(aes(x=1.1, y=0.06), colour="red")
```

Warning in geom_point(aes(x = 1.1, y = 0.06), colour = "red"): All aesthetics have length 1, i Please consider using `annotate()` or provide this layer with data containing a single row.



```
Datos <- as.timeSeries(Datos)
Datos <- portfolioData(data=Datos, spec=portfolioSpec())
getStatistics(Datos)
```

\$mean

	R.C	R.M	R.P	R.D	R.S
	0.099810534	0.068728418	0.034220986	-0.001765793	0.108976913

\$Cov

	R.C	R.M	R.P	R.D	R.S
R.C	2.1486098	1.200212	0.9140233	0.9237187	1.3515339
R.M	1.2002122	3.435939	1.0450716	2.2879038	1.9594761
R.P	0.9140233	1.045072	1.6699095	0.7710093	0.3741879
R.D	0.9237187	2.287904	0.7710093	4.2788822	2.2418133
R.S	1.3515339	1.959476	0.3741879	2.2418133	9.2172984

\$estimator

[1] "covEstimator"

\$mu

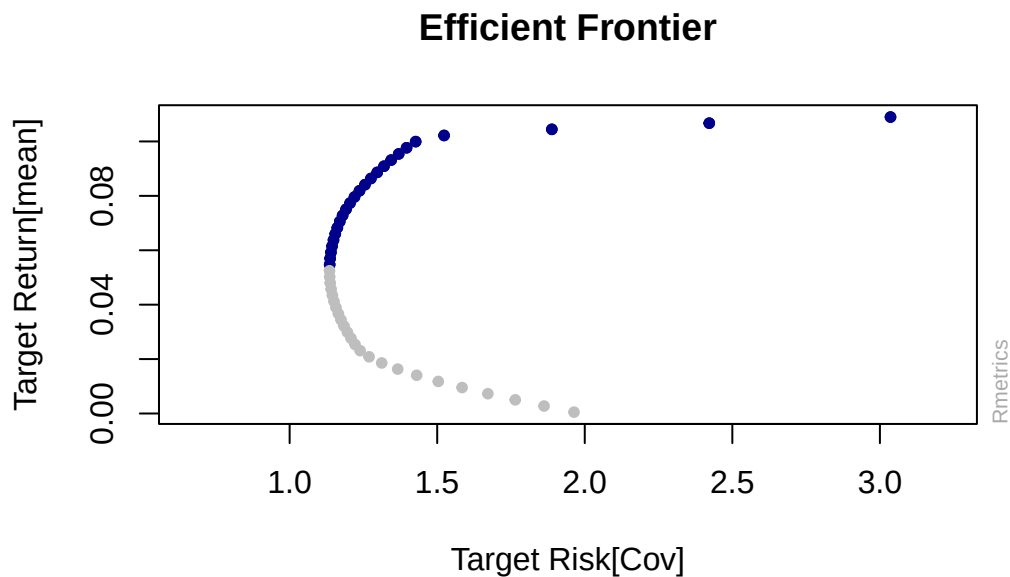
	R.C	R.M	R.P	R.D	R.S
	0.099810534	0.068728418	0.034220986	-0.001765793	0.108976913

\$Sigma

	R.C	R.M	R.P	R.D	R.S
R.C	2.1486098	1.200212	0.9140233	0.9237187	1.3515339
R.M	1.2002122	3.435939	1.0450716	2.2879038	1.9594761
R.P	0.9140233	1.045072	1.6699095	0.7710093	0.3741879
R.D	0.9237187	2.287904	0.7710093	4.2788822	2.2418133
R.S	1.3515339	1.959476	0.3741879	2.2418133	9.2172984

```
#Frontera eficiente
frontera <- portfolioFrontier(Datos)

frontierPlot(frontera, cex=1, pch=20, col=c("blue4", "gray"))
```



```
#montecarlo
frontierPlot(frontera, cex = 1, pch = 20,

col = c("blue4", "gray"))

#SIMULACIÓN DE PORTAFOLIOS (Portafolios con diferentes pesos)
monteCarloPoints(frontera, mcSteps = 1000,
```

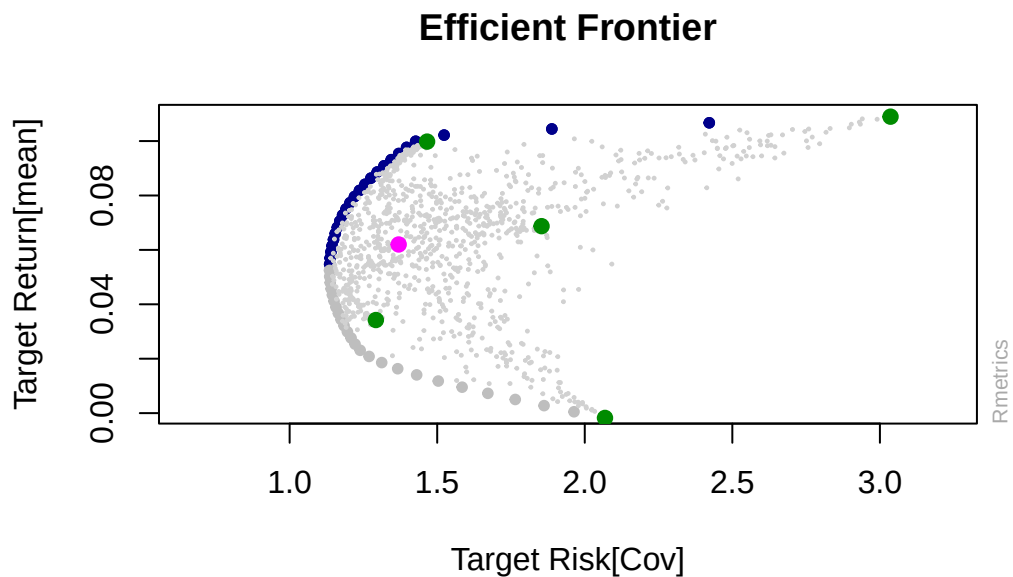
```

pch = 19, col = "gray82", cex = 0.2)
#PORTAFOLIOS CON UN SOLO ACTIVO (1,0,0,0,0), (0,1,0,0,0) , (0,0,1,0,0) , (0,0,0,1,0) y (0,0,0,0,1)
singleAssetPoints(frontera,

col = "green4", pch = 19, cex = 1)
#PORTAFOLIO CON PESOS IGUALES (1/5, 1/5, 1/5, 1/5, 1/5)
equalWeightsPoints(frontera,

cex = 1, col = "magenta", pch = 19)

```



```

#min global
min.var <- efficientPortfolio(data = Datos, constraints = "LongOnly")

min.var

```

Title:
 MV Efficient Portfolio
 Estimator: covEstimator
 Solver: solveRquadprog
 Optimize: minRisk
 Constraints: LongOnly

Portfolio Weights:

R.C	R.M	R.P	R.D	R.S
0.2843	0.0070	0.5492	0.1125	0.0470

Covariance Risk Budgets:

R.C	R.M	R.P	R.D	R.S
0.2843	0.0070	0.5492	0.1125	0.0470

Target Returns and Risks:

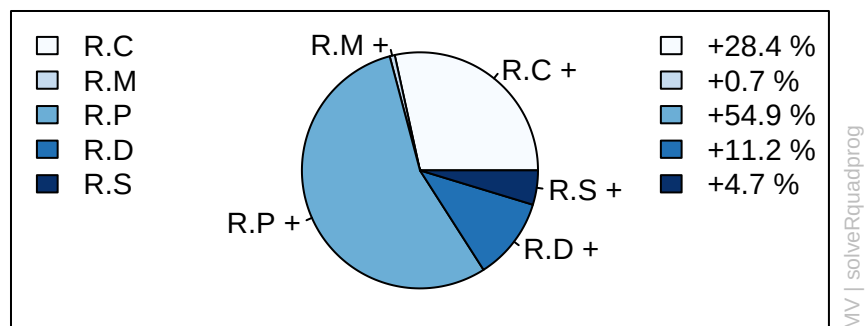
mean	Cov	CVaR	VaR
0.0526	1.1352	2.6456	1.6207

Description:

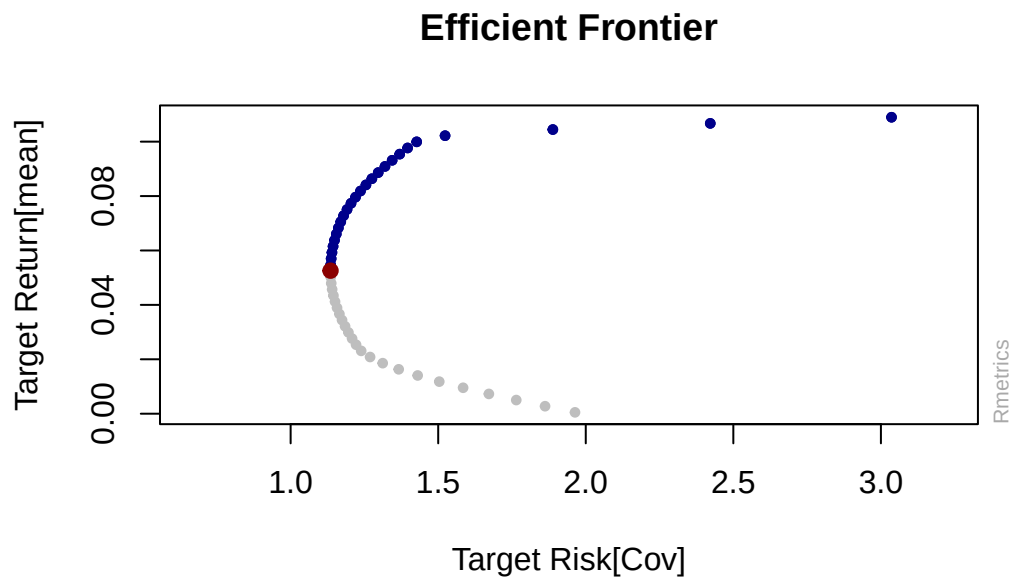
Sun May 11 12:02:52 2025 by user: Blanca

```
weightsPie(min.var)
```

Weights



```
frontierPlot(frontera, cex=0.6, pch=19, col=c("blue4", "gray"))  
minvariancePoints(min.var, col="red4", pch=19, cex=1)
```



```
#portafolio con retno esperado
mu.bar <- 0.1 # con un mu.bar>0.1 se va todo a CERO
esp.cartera <- portfolioSpec()
setTargetReturn(esp.cartera) = mu.bar
por.efi <- efficientPortfolio(data = Datos, esp.cartera,
constraints = "LongOnly")

por.efi
```

Title:
 MV Efficient Portfolio
 Estimator: covEstimator
 Solver: solveRquadprog
 Optimize: minRisk
 Constraints: LongOnly

Portfolio Weights:

R.C	R.M	R.P	R.D	R.S
0.8627	0.0266	0.0000	0.0000	0.1108

Covariance Risk Budgets:

R.C	R.M	R.P	R.D	R.S
0.8609	0.0175	0.0000	0.0000	0.1216

Target Returns and Risks:

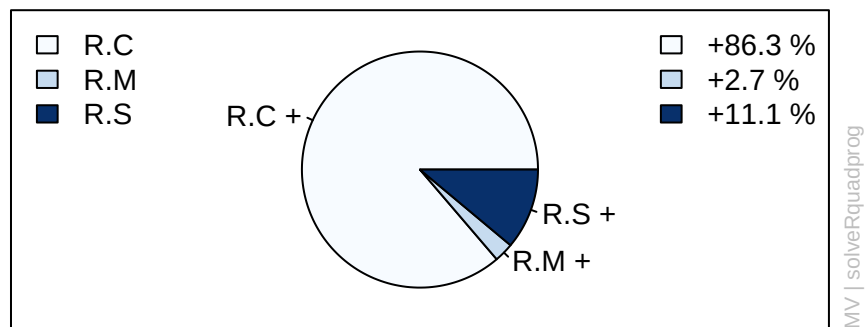
mean	Cov	CVaR	VaR
0.1000	1.4281	3.3484	2.1131

Description:

Sun May 11 12:02:52 2025 by user: Blanca

```
weightsPie(por.efi)
```

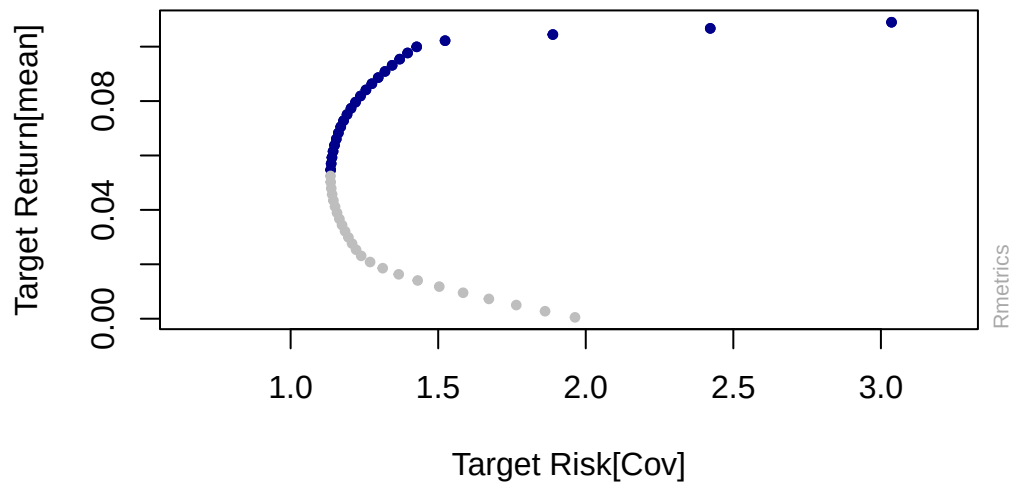
Weights



```
sigma<- as.numeric(getTargetRisk(por.efi)["Cov"])
mu <- as.numeric(getTargetRisk(por.efi))[1]

frontierPlot(frontera, cex=0.6, pch=19, col=c("blue4", "gray"))
points(x=sigma, y=mu, pch=4, cex=1)
```

Efficient Frontier



```
#portafolio tangente
R.f <- 10 / 365
tg.p <- portfolioSpec()
setRiskFreeRate(tg.p) <- R.f

tgPortafolio <- tangencyPortfolio(Datos, spec = tg.p,
constraints = "LongOnly")

#PUNTO PARA GRÁFICAR
sigma <- as.numeric(getTargetRisk(tgPortafolio)["Cov"])
mu <- as.numeric(getTargetReturn(tgPortafolio))[1]
tgPortafolio
```

Title:

MV Tangency Portfolio

Estimator: covEstimator

Solver: solveRquadprog

Optimize: minRisk

Constraints: LongOnly

Portfolio Weights:

R.C	R.M	R.P	R.D	R.S
0.878	0.000	0.000	0.000	0.122

Covariance Risk Budgets:

R.C	R.M	R.P	R.D	R.S
0.8647	0.0000	0.0000	0.0000	0.1353

Target Returns and Risks:

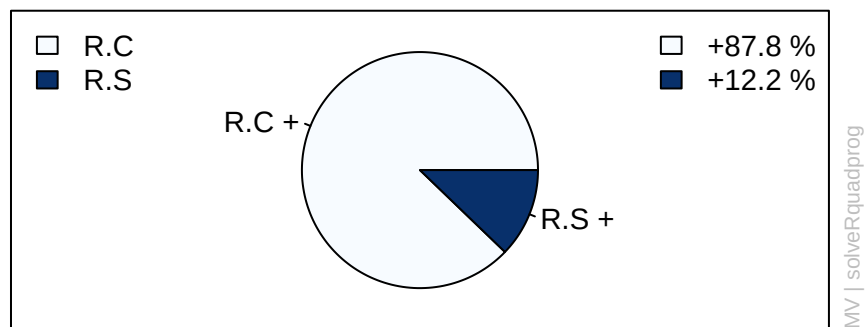
mean	Cov	CVaR	VaR
0.1009	1.4433	3.3767	2.1299

Description:

Sun May 11 12:02:52 2025 by user: Blanca

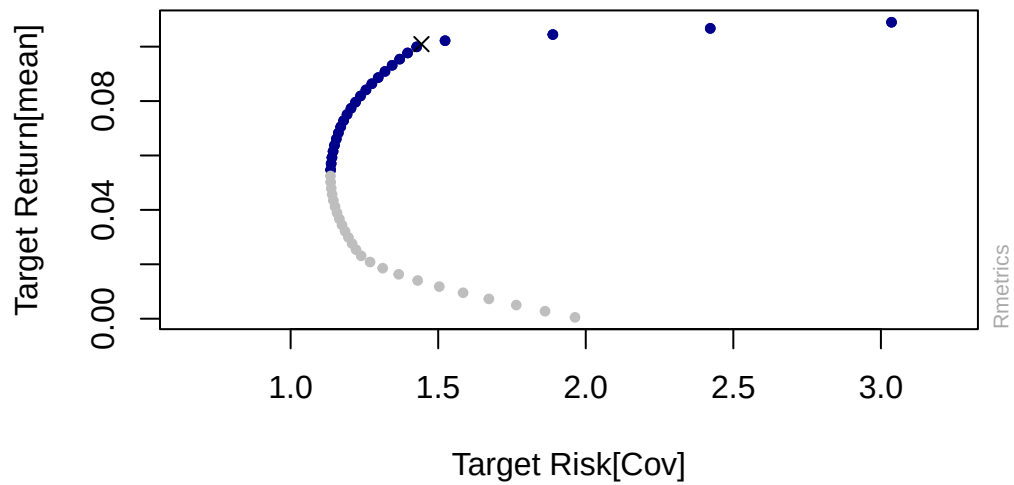
```
weightsPie(tgPortafolio)
```

Weights



```
frontierPlot(frontera, cex=0.6, pch=19, col=c("blue4", "gray"))
points(x=sigma, y=mu, pch=4, cex=1)
```

Efficient Frontier



```
mu.tan<- getTargetReturn(tgPortafolio)[1]
s.tan<-getTargetRisk(tgPortafolio)[1]
(mu.tan-R.f)/s.tan
```

```
mean
0.0509475
```

```
frontierPlot(frontera, pch=19, cex=0.6, col=c("blue4", "green4"))
tangencyLines(tgPortafolio, return="mu", col="red4")
points(x=sigma, y=mu, pch=4, cex=1)
```


Efficient Frontier

